**SIMATS ENGINEERING**

**CO1 - ASSESSMENT TOOL**

**3 EXPERIMENTS**

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|---|---|---|
| **Name of the Student** | **N.Jeevan sai** | |
| **Reg. No** | **192525169** | |
| | | |

**COURSE NAME: Deep Learning** **COURSE CODE: MLA0403 FACULTY : Dr.Arul Raj A.M**

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**COURSE OUTCOME COVERED**

**CO 1:** Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

**CO1 Weightage: 15%**

**Assessment Tool Weightage: 35%**

**Duration: 30 minutes**

**Marks: 50**

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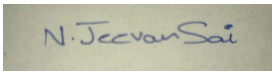
**Code of Conduct:**

I **N.Jeevan sai /192525169** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.




**Signature of the student:**

**Name of the student: N.Jeevan sai**

**QUESTIONS:10** **Total Marks:50**

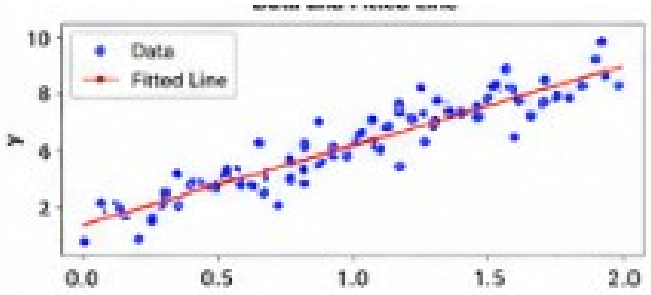
**CO1- AT3**
**Experiments Questions**

## 1. Learning Algorithms

Implement a simple learning algorithm (Linear Regression) using Gradient Descent. Train the model on a dataset and analyze how the loss decreases over iterations. Plot the learning curve.

**Answer :**

**Generate a dataset with input *X*and output *Y*.**




- Initialize weights randomly.
- Update weights using Gradient Descent.
- Compute Mean Squared Error (MSE) after each iteration.
- Plot Loss vs Iterations.
- Observation: Loss decreases gradually and converges to a minimum, indicating successful learning.

## 2. Maximum Likelihood Estimation (MLE)

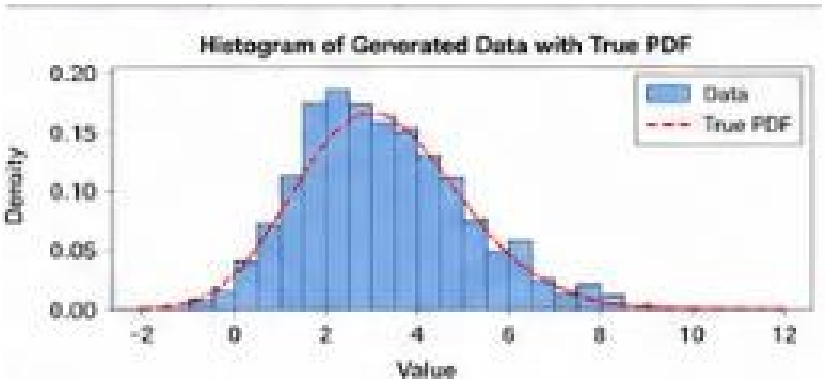
Generate synthetic data from a normal distribution and use MLE to estimate the mean and variance. Compare estimated values with actual parameters.

**Answer :**

- Generate synthetic data from a normal distribution $N(\mu, \sigma^2)$.
- Estimate

$$\hat{\mu} = \frac{1}{n}\sum x_i$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum (x_i - \hat{\mu})^2$$



- Compare estimated values with actual parameters.

- Observation: Estimated mean and variance are very close to the true values as sample size increases.

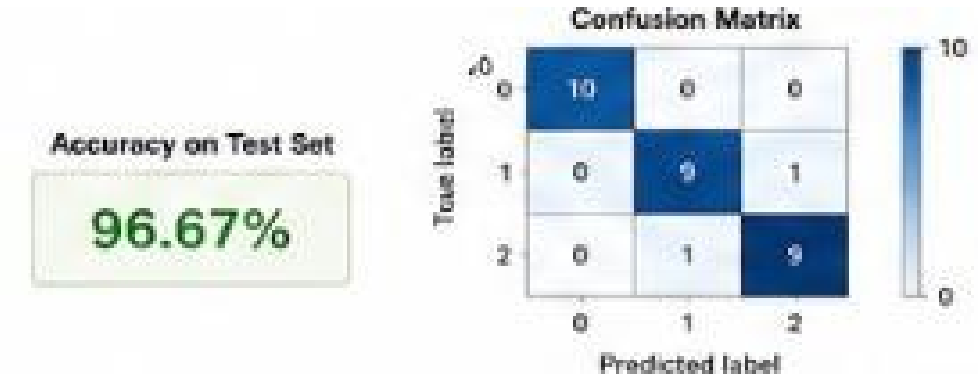
## 3. Building Machine Learning Algorithms

Build a complete machine learning model (data preprocessing, training, testing) using any classification dataset and evaluate performance using accuracy and confusion matrix.

**Answer :**

Steps:

1. Load dataset (e.g., Iris).
2. Handle missing values.
3. Normalize features.
4. Split into train and test sets.
5. Train a classifier (Logistic Regression/Decision Tree).
6. Predict test samples.
7. Evaluate using:
    - Accuracy
    - Confusion Matrix



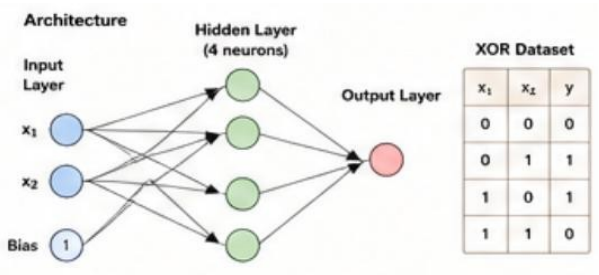

Observation:
High accuracy and diagonal confusion matrix indicate good classification performance.

## 4. Neural Networks

**Q4.** Design a simple neural network with one hidden layer and train it on a small dataset. Analyze how the network learns non-linear patterns.

**Answer :**

1. Build a neural network with:
2. Input layer
3. One hidden layer
4. Output layer
5. Train on XOR dataset.
6. The hidden layer enables learning of non-linear decision boundaries.



- Loss decreases over epochs, showing successful learning.

## 5. Artificial Neurons

Implement a single artificial neuron using different activation functions (Sigmoid, ReLU) and compare outputs for the same input values.

**Answer :**

Implement a single neuron using:

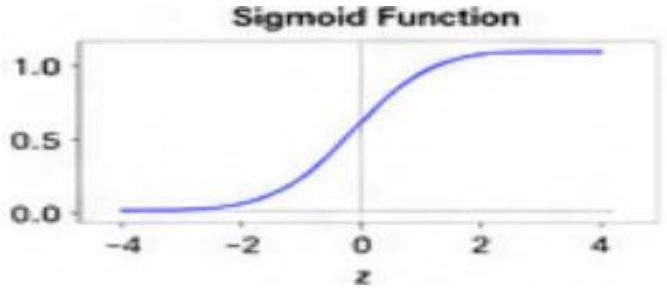
Sigmoid

$$f(x) = \frac{1}{1 + e^{-x}}$$

ReLU

$$f(x) = \max(0, x)$$

Comparison




| **Sigmoid** | **ReLU** |
|---|---|
| Output between 0 and 1 | Output ≥0 |
| Smooth curve | Fast computation |
| Suffers vanishing gradient | Preferred in deep learning |

## 6. Perceptron and Multilayer Perceptron (MLP)

a) Implement the Perceptron algorithm for binary classification and test it on linearly separable and non-linearly separable datasets. Analyze the results.

**Answer :**

(a)
Perceptron
Answer:

- Train on a linearly separable dataset.
- Perceptron converges successfully.
- On non-linearly separable datasets (e.g., XOR), it fails to converge.

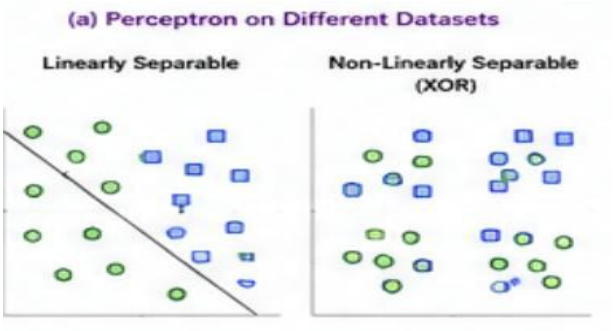
(b) MLP
Answer:

- Train an MLP with one hidden layer.
- Compare with

Perceptron. Observation

- Perceptron works only for linear problems.

- MLP successfully learns non-linear patterns and achieves higher accuracy.




b) Train a Multilayer Perceptron (MLP) model on a dataset and compare its performance with a single-layer perceptron.

**Answer :**

(b) MLP
Answer:

- Train an MLP with one hidden layer.
- Compare with
  Perceptron. Observation
- Perceptron works only for linear problems.
- MLP successfully learns non-linear patterns and achieves higher accuracy.

## 7. Forward Propagation and Backpropagation Algorithm

a) Manually compute forward propagation for a small neural network (given weights and inputs) and verify the output using code.
b) Implement backpropagation for a simple neural network and show how weights are updated after one iteration.

**Answer :**

**(a) Forward Propagation**

- Given inputs, weights and bias:
  - Compute hidden layer outputs.
  - Apply activation function.
  - Compute final output.
- Verify calculations using Python.

**(b)**

**Backpropagation**

**Answer:**

- Compute output error.
- Calculate gradients.
- Update weights using:

## 8. Gradient Descent and Stochastic Gradient Descent (SGD)

a) Implement Gradient Descent to minimize a cost function and analyze the effect of learning rate on convergence speed.

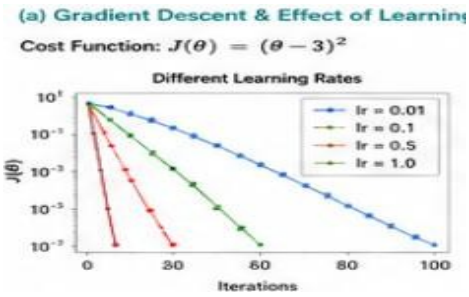
**Answer :**

**(a) Gradient Descent**

- Minimize the cost function.
- Compare learning rates.

**Observation:**

- Small learning rate → slow convergence.
- Very large learning rate → divergence.
- Moderate learning rate → fastest stable convergence.




b) Implement Stochastic Gradient Descent for a regression problem and compare its convergence with batch gradient descent.

**Answer :**

**(b) Stochastic Gradient Descent**

- Updates parameters after every training sample.
- Converges faster than Batch Gradient Descent.
- Loss fluctuates due to noisy updates.

## 9. Mini-Batch Gradient Descent

Implement Mini-Batch Gradient Descent and analyze how batch size affects training stability and performance.

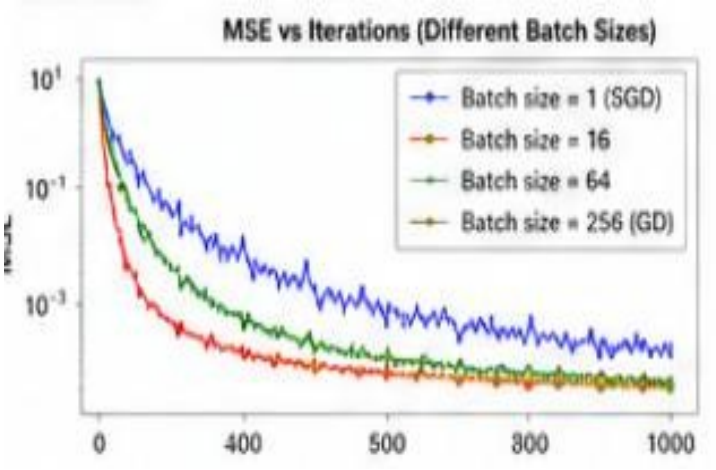
**Answer :**

Train with different batch sizes.

Observation

- Batch size = 1 (SGD): Fast but noisy.
- Batch size = 16–64: Best balance.
- Batch size = Full dataset: Stable but slow.

Mini-Batch GD is generally preferred because it balances speed and stability.




## 10. Curse of Dimensionality

Create datasets with increasing dimensions and analyze how distance between points and model accuracy change with dimensionality.

**Answer :**

Create datasets with increasing dimensions.

Analyze:

- Average distance between points.
- Classification accuracy.

Observation

- Distances become increasingly similar.
- Distance-based algorithms become less effective.
- Model accuracy decreases due to sparse data and overfitting.

Solutions:

- PCA
- Feature Selection
- L1/L2 Regularization
- Autoencoders